

Echocardiographic Markers in the Diagnosis of Cardiac Masses



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Background: The echocardiographic parameters required for a comprehensive assessment of cardiac masses (CMs) are still largely unknown. The aim of this study was to identify and integrate the echocardiographic features of CMs that can accurately predict malignancy.

Methods: An observational cohort study was conducted among 286 consecutive patients who underwent standard echocardiographic assessment for suspected CM at Bologna University Hospital between 2004 and 2022. A definitive diagnosis was achieved by histologic examination or, in the case of cardiac thrombi, with radiologic evidence of thrombus resolution after appropriate anticoagulant treatment. Logistic and multivariable regression analysis was performed to confirm the ability of six echocardiographic parameters to discriminate malignant from benign masses. The unweighted count of these parameters was used as a numeric score, ranging from 0 to 6, with a cutoff of ≥ 3 balancing sensitivity and specificity with respect to the histologic diagnosis of malignancy. Classification tree analysis was used to determine the ability of echocardiographic parameters to discriminate subgroups of patients with differential risk for malignancy.

Results: Benign masses were more frequently pedunculated, mobile, and adherent to the interatrial septum ($P < .001$). Malignant masses showed a greater diameter and exhibited a higher frequency of irregular margins, an inhomogeneous appearance, sessile implantation, polylobate shape, and pericardial effusion ($P < .001$). Infiltration, moderate to severe pericardial effusion, nonleft localization, sessile implantation, polylobate shape, and inhomogeneity were confirmed to be independent predictors of malignancy in both univariate and multivariable models. The predictive ability of the unweighted score of ≥ 3 was very high (>0.90) and similar to that of the previously published weighted score. Classification tree analysis generated an algorithm in which infiltration was the best discriminator of malignancy, followed by nonleft localization and sessile implantation. The percentage correctly classified by classification tree analysis as malignant was 87.5%. Agreement between observer readings and CM histology ranged between 85.1% and 91.5%. The presence of at least three echocardiographic parameters was associated with lower survival.

Conclusions: In the approach to CMs, some echocardiographic parameters can serve as markers to accurately predict malignancy, thereby informing the need for second-level investigations and minimizing the diagnostic delay in such a complex clinical scenario. (J Am Soc Echocardiogr 2023;36:464-73.)

Keywords: Cardiac masses, Echocardiography, Pseudotumors, Primary benign tumors, Primary malignant tumors, Secondary malignant tumors, Diagnosis

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Abbreviations

AUC	= Area under the curve
CCT	= Cardiac computed tomography
CM	= Cardiac mass
CMR	= Cardiac magnetic resonance
CTA	= Classification tree analysis
LV	= Left ventricular

Cardiac masses (CMs) are an extremely heterogeneous clinical entity, including benign and malignant lesions.^{1,2} The reported prevalence has increased over the past decade, most likely related to the wider availability of second- and third-level imaging techniques. Nevertheless, in current clinical practice, CM diagnosis remains challenging, and targeted diagnostic protocols are warranted as the correct identification of masses' nature (i.e., benign vs malignant) is

essential to guide the proper treatment.³⁻⁶ Although symptoms and signs of CMs are often nonspecific, some clinical and imaging findings may be crucial to guide the diagnostic workup.^{7,8} The gold standard for the diagnosis of CM had traditionally been endomyocardial biopsy, but its invasive nature, costs, and technical complexity reduced its use in daily practice. Thus, the demand for cardiac imaging has increased, reserving biopsy for cases with inconclusive noninvasive investigations. Echocardiography is part of a routine cardiologic evaluation and represents the first-line imaging approach to CMs.^{9,10} In fact, echocardiographic parameters provide crucial information, including the location, size, morphology, mobility, pericardial involvement, and hemodynamic impact of a mass. Unfortunately, echocardiographic imaging alone is often insufficient to accurately define CMs' characteristics because of poor acoustic windows, operator dependence, and artifacts that may occasionally be misleading. Therefore, a stepwise imaging approach with cardiac magnetic resonance (CMR), cardiac computed tomography (CCT), and, eventually, positron emission tomography is usually needed.¹¹⁻¹⁴ However, for the assessment of CMs, these techniques still lack standardized protocols, and there is no recommended multimodality diagnostic algorithm.¹⁵ In addition, although these radiologic tools are widely available at tertiary centers, smaller facilities still rely on simpler methods, so when a CM is suspected or identified, a thorough echocardiographic evaluation can reveal important information. Nevertheless, the echocardiographic parameters required for a comprehensive assessment of patients with CMs are still largely unknown, even though some features have been related to masses' nature (benign vs malignant) in a few retrospective studies with limited sample sizes.¹⁶⁻¹⁸ In a recent study, we used a weighted diagnostic echocardiographic score to predict the malignant nature of CMs and consequently refer patients to second-level imaging techniques.¹⁹ In the present work, we tested whether the simple sum of echocardiographic features could accurately suggest masses' nature, and we looked for a classification algorithm to discriminate benign versus malignant masses.

METHODS

Study Population

This observational cohort study included all consecutive patients who underwent noninvasive imaging investigations (transthoracic echocardiography, CCT, and CMR) for suspected CMs at University Hospital Policlinico Sant'Orsola Malpighi in Bologna, Italy, from January 2004 to September 2022. A definitive diagnosis

was achieved by the histologic examination of biopsy or surgical samples or, in the case of cardiac thrombi, by radiologic evidence of thrombus resolution after adequate anticoagulant treatment. After diagnostic workup, CMs were classified as benign or malignant. Benign masses included pseudotumors and primary cardiac benign tumors, whereas malignant masses included primary cardiac malignant and secondary tumors.^{1,2} All cases were classified according to the World Health Organization 2015 classification of tumors of the heart and pericardium¹; sarcomas were graded according to the Fédération Nationale des Centres de Lutte Contre le Cancer system.^{20,21} Pseudotumors, which represent an extremely heterogeneous group, were defined as lesions not originating from neoplastic transformations of specific cell types.²² Normal anatomic variants were excluded because of the inability to obtain a histologic examination. In patients with cardiac lesions and concomitant clinical suspicion of infective endocarditis, all imaging and laboratory examinations were performed to eventually diagnose infective endocarditis according to guidelines.²³ Subsequently, only patients with CM who did not meet the diagnostic criteria for infective endocarditis were included in our registry. The study flowchart is shown in [Supplemental Figure 1](#). All patients were managed according to the Declaration of Helsinki and provided informed consent for the anonymous publication of scientific data. The study protocol was approved by the local ethics committee (registration number 102/2017/Oss/AOUBo).

Data Collection and Outcomes

For each patient, demographic and clinical data were collected, including age, sex, anthropometric data, cardiovascular risk factors, history of comorbidities, and first admission diagnosis. All patients underwent complete diagnostic workup, including clinical evaluation and laboratory testing (with specific examinations according to the clinical scenario). All subjects were followed after the index presentation, and clinical follow-up data were obtained from clinical visits or telephone interviews.

Echocardiography

All patients were evaluated using two-dimensional echocardiography using a high-quality ultrasound machine (iE33 or Epiq 7C; Philips Healthcare) with a transthoracic echocardiographic transducer (S5-1). Echocardiographic examinations were performed following the recommendations of the American Society of Echocardiography and the European Association of Cardiovascular Imaging with patients in the left lateral recumbent position with the use of standard views.²⁴⁻²⁷ At least three consecutive beats were recorded for each view, and all images were stored for offline analysis (IntelliSpace, Philips Healthcare). The recorded images were analyzed offline by an expert echocardiography cardiologist (M.G.), with more than 10 years' experience in cardiac imaging, blinded to clinical information and CM histology. The cardiac imager analyzed all echocardiograms using a prespecified worksheet to mark the presence or absence of the following parameters: location (left or right, atrium or ventricle, pericardium, great vessels), site of attachment (interatrial or interventricular septum or roof or side wall of the atrium, ventricular free wall), dimension, shape (regular or irregular), margins (well defined or irregular [if more than 50% of the border was clearly demarcated]), mass characteristics (sessile [attached directly by the base and not raised upon a stalk], pedunculated [raised upon a stalk], or polylobate [having two or more lobes]), mobility, infiltration

HIGHLIGHTS

- The echo parameters required for a comprehensive CM assessment are still unknown.
- Six prespecified echo features of CM can rapidly raise the suspicion of malignancy.
- Combining three or more parameters, diagnostic accuracy reaches approximately 90%.
- An algorithm was generated that correctly classified 87.5% of malignancies.
- These echocardiographic features also proved to have prognostic validity.

(defined as disruption of neighboring tissue and extension of the mass across the pericardium into myocardium, with interruption of epicardial and endocardial contours or, alternatively, by the presence of at least one of the following echocardiographic features: [i] evidence of a different reflectivity compared with the normal myocardium, as infiltrating masses usually have a peculiar, granular echocardiographic texture; [ii] increased thickness in comparison with the adjacent myocardial segments; and [iii] hypokinesia or akinesia of a focal myocardial area compared with closest cardiac segments in absence of coronary distribution that could lead to the suspicion of ischemic etiology),²⁸ pericardial effusion (defined as a fluid accumulation between the two pericardial layers), and echogenicity pattern (hypo-, iso-, or hyperechogenic compared with normal myocardium). Left ventricular (LV) end-diastolic diameter and volume were calculated according to the American Society of Echocardiography and European Association of Cardiovascular Imaging recommendations.^{26,27} LV ejection fraction was calculated using Simpson's biplane method. LV diastolic function was assessed using the recommended algorithm proposed, including E wave, e' velocities, E/e' ratio, left atrial volume index, and tricuspid regurgitation velocity. Last, we evaluated the diagnostic agreement of echocardiography with second-level imaging techniques, specifically CMR and CCT.

Statistical Analyses

Data are reported as mean ± SD or median (interquartile range). The Pearson χ^2 test was used to compare categorical variables among groups. The normality of the distribution of continuous variables was assessed using the Shapiro-Wilk test, and equality of variance was tested between groups using the Levene test. In case of departure from normality, nonparametric tests were used (Mann-Whitney *U* test or Kruskal-Wallis test), while the Welch test was used when variances were unequal between groups. The association between echocardiographic parameters and the probability of malignancy was assessed using univariate logistic regression analyses. Parameters significantly associated with malignancy were entered in a multivariable logistic regression model. Parameters retained in the multivariable model were used to build for each patient an unweighted score (the count of the echocardiography parameters coded as 1) and a weighted score, according to our previous publication.¹⁹ The best cutoff of the count to predict malignancy was determined as the one that maximized the Youden index (sensitivity + specificity – 1).

Classification tree analysis (CTA) was used to determine the ability of echocardiographic parameters to discriminate subgroups of patients with differential risk for malignancy. In contrast to traditional statistical models, this nonparametric technique simultaneously examines interactions between categorical (or continuous) variables to create a decision tree that does not rely on assumptions about linear relationships between dependent and independent variables. CTA is represented graphically as an inverted tree. Beginning with a root node, which includes all cases, the tree branches iteratively. The best discriminating predictor is selected first, and then subsequent predictors are entered into the procedure if they contribute significantly to subtyping cases that are homogeneous groups in terms of the value of the dependent variable. The homogeneity of each node was measured using the Gini index. The percentage of cases correctly classified was used to determine the goodness of fit of the multivariable models and of the CTA model. Last, the echocardiographic unweighted count was associated using Kaplan-Meier curve analysis to determine its ability to correlate with patient survival. The significance level was set to *P* < .05. Statistical analyses were carried out using SPSS Statistics version 28.0.1.1. (IBM) and Stata version 17 (StataCorp).

Table 1 Histologic characterization of benign and malignant masses

CMs (n = 286)			
Benign masses (n = 212)		Malignant masses (n = 74)	
Primary benign tumors (n = 145 [50.7%])		Primary malignant tumors (n = 21 [7.3%])	
Myxoma	112 (77.2)	Sarcoma	17 (81)
Fibroelastoma	22 (15.2)	Lymphoma	3 (14)
Lipoma	5 (3.4)	Mesothelioma	1 (5)
Fibroma	3 (2.1)		
Paraganglioma	2 (1.4)		
Hamartoma	1 (0.7)		
Pseudotumors (n = 67 [23.4%])		Metastasis (n = 53 [18.6%])	
Thrombus	35 (52.2)	Lymphoma	21 (39.6)
Cyst	8 (12)	Sarcoma	7 (13.2)
Valvular nodule	8 (12)	Melanoma	5 (9.5)
Lipomatosis	6 (8.9)	Hepatocellular carcinoma	5 (9.5)
Reactive inflammatory process	6 (8.9)	Colon carcinoma	4 (7.5)
Calcification	3 (4.5)	Renal and urological tumor	4 (7.5)
Cystic atrioventricular node tumor	1 (1.5)	Lung carcinoma	4 (7.5)
		Gynecological tumor	2 (3.8)
		Plasmacytoma	1 (1.9)

Data are expressed as number (percentage).

Table 2 Comparison of demographic, medical history, clinical, and laboratory features between benign and malignant CMs

	Total population (n = 286)	Benign CMs (n = 212)	Malignant CMs (n = 74)	P
Gender, male	131 (45.8)	83 (39.2)	48 (64.9)	<.001
Age, y	60.1 ± 15.6	60.4 ± 15.1	59 ± 17	.612
Age groups				.606
21-40 y	38 (13.3)	25 (11.8)	13 (17.6)	
41-60 y	91 (31.8)	66 (31.4)	25 (33.8)	
61-80 y	138 (48.3)	107 (50.5)	31 (41.9)	
81-100 y	18 (6.3)	13 (6.1)	5 (6.8)	
BMI, kg/m ²	24.1 ± 4.3	23.7 ± 4.1	25.8 ± 4.6	.089
Cardiovascular risk factors				
Smoking habit	142 (49.7)	102 (48.1)	40 (54.1)	.379
Hypertension	159 (55.6)	122 (57.5)	37 (50)	.261
Dyslipidemia	124 (43.4)	98 (46.2)	26 (35.1)	.097
T2DM	41 (14.3)	31 (14.6)	10 (13.5)	.815
Medical history				
Congestive HF	34 (11.9)	25 (11.8)	9 (12.2)	.933
Prior stroke	64 (22.4)	51 (24.1)	13 (17.6)	.249
History of neoplasia	93 (32.5)	65 (30.7)	28 (37.8)	.256
CHA ₂ DS ₂ -VASc	3 (1-4)	3 (1-4)	2 (1-3)	.018
Histology and location				
Pseudotumor	67 (23.4)	67 (31.6)	0 (0)	<.001
Primary benign	145 (50.7)	145 (68.4)	0 (0)	
Primary malignant	21 (7.3)	0 (0)	21 (28.4)	
Metastasis	53 (18.5)	0 (0)	53 (71.6)	
Right cardiac chambers	85 (29.7)	43 (20.3)	42 (56.8)	<.001
Left cardiac chambers	167 (58.4)	154 (72.6)	13 (17.6)	
Pericardium	24 (8.4)	14 (6.6)	10 (13.5)	
Great vessels	10 (3.5)	1 (0.5)	9 (12.2)	
Clinical features				
Incidental diagnosis	128 (44.8)	110 (51.9)	18 (24.3)	<.001
Dyspnea	112 (39.2)	68 (32.1)	44 (59.5)	<.001
NYHA class				<.001
I or II	219 (76.6)	176 (83)	43 (58.1)	
III or IV	67 (23.4)	36 (17)	31 (41.9)	
Chest pain	42 (14.7)	29 (13.7)	13 (17.6)	.416
Palpitations	38 (13.3)	30 (14.2)	8 (10.8)	.466
Peripheral embolism	40 (14)	35 (16.5)	5 (6.8)	.037
Pulmonary embolism	27 (9.4)	11 (5.2)	16 (21.6)	<.001
Laboratory parameters				
Creatinine, mg/dL	1.05 ± 1	1.13 ± 1.1	0.80 ± 0.3	.324
GFR, mL/min	87 (72-98)	83 (70-98)	92 (83-110)	.058
Hb, g/dL	12.4 ± 1.7	12.6 ± 1.6	12.1 ± 1.9	.266

(Continued)

Table 2 (Continued)

	Total population (n = 286)	Benign CMs (n = 212)	Malignant CMs (n = 74)	P
WBC count, n/mm ³	7,540 (6,190-9,950)	7,390 (6,120-8,980)	9,085 (6,622-10,895)	.004
CRP, mg/dL	3.7 ± 3	2.5 ± 2	5.8 ± 3.4	<.001

BMI, Body mass index; CRP, C-reactive protein; GFR, glomerular filtration rate; Hb, hemoglobin; HF, heart failure; NYHA, New York Heart Association; T2DM, type 2 diabetes mellitus; WBC, white blood cell.

Continuous variables are expressed as mean ± SD or as median (interquartile range) as appropriate and categorical variables as number (percentage).

RESULTS

A total of 369 patients included in the Bologna Cardiac Masses Registry underwent complete recorded transthoracic and/or transesophageal echocardiographic evaluation from January 2004 to September 2022. As shown in the study flowchart (Supplemental Figure 1), we excluded 83 cases with unavailable histology, extracardiac masses as confirmed by second-level investigations, poor acoustic windows, and masses that were not visualized on echocardiography. The latter included nine pericardial masses, seven involving the pulmonary arteries, and 11 small lesions, of which three were incidentally diagnosed during valve surgery. As a result, our study population consisted of 286 patients. Therefore, the sensitivity of diagnostic echocardiography in detecting a CM was 91.4% (286 of 313). On the basis of histopathologic examination, we identified 212 patients (74.1%) with benign CMs and 74 (25.9%) with malignancies (Table 1). As for the 35 cardiac thrombi, the histology was obtained by surgery or radiologic resolution after adequate anticoagulation treatment. In fact, 33 of our thrombi resolved after adequate anticoagulant treatment, while in two cases, the diagnosis was reached after surgery. In two additional circumstances, the suspected thrombi did not resolve after anticoagulation, and the masses were subsequently reevaluated and reclassified, one as a myxoma and one as a cyst. Each patient enrolled completed follow-up, with a median follow-up time of 22 months and a maximum of 180 months (15 years).

Benign versus Malignant Masses: Clinical Presentation

Baseline characteristics, cardiovascular risk factors, comorbidities, and clinical presentation are reported in Table 2. Age at presentation, body mass index, main cardiovascular risk factors, and comorbidities were similar in the two groups. Patients with malignant masses were more frequently male, presenting with dyspnea (mostly New York Heart Association functional class III or IV) and with a lower rate of incidental diagnosis than patients with benign masses ($P < .001$). Benign formations were often located in the left heart chambers, whereas malignancies were usually detected on the right side, in the pericardium or the pulmonary arteries ($P < .001$). In consequence, patients with benign masses exhibited a greater peripheral embolization, whereas malignant masses were more often associated with pulmonary embolisms ($P < .05$ for both). Moreover, malignant CMs exhibited a higher level of inflammation as expressed by C-reactive protein and total white blood cell values ($P < .01$).

Benign versus Malignant Masses: Echocardiographic Features

Echocardiographic parameters significantly differed between benign and malignant masses (Table 3). Benign masses were more frequently pedunculated, mobile, and adherent to the interatrial septum ($P < .001$ for all). On the other hand, malignant masses showed a greater diameter (>30 mm was significantly associated with malignancies) and exhibited a higher frequency of irregular margins, inhomogeneous appearance, sessile implantation, and polylobate shape ($P < .01$ for all). Finally, compared with benign ones, malignant masses more frequently presented with pericardial effusion ($P < .001$). When benign CMs presented with pericardial effusion, the effusion grade was mild. LV ejection fraction and LV end-diastolic volume were similar between the two study groups.

Score-Based versus CTA-Based Diagnostic Approach

Infiltration, moderate to severe pericardial effusion, nonleft localization, sessile implantation, polylobate shape, and inhomogeneity were confirmed to be independent predictors of malignancy in both univariate and multivariate models (Figure 1 and Supplemental Table 1). An unweighted score of these six positive parameters, ranging from 0 to 6, was computed for each patient to denote the echocardiographic risk for malignancy. A cutoff of three or more positive parameters was obtained on the basis of the Youden index, which maximized sensitivity and specificity with respect to the histologic diagnosis of malignancy. Case examples are provided in Figure 2 and Videos 1 to 6 (available at www.onlinejase.com). The predictive ability of the unweighted count was compared with the weighted score derived in our previous publication.¹⁹ These two scores (weighted and unweighted) were compared using receiver operating characteristic analysis and proved to have a very high area under the curve (AUC) (>0.90) and similar diagnostic accuracy, with a modest nonsignificant advantage of the weighted score ($P = .065$; Figure 3). Last, CTA generated an algorithm in which infiltration was the best discriminator of malignancy, with a positive predictive value of 82.5%. Among those without infiltration, nonleft localization proved to be able to further distinguish malignant cases. Then, among those without nonleft localization, sessile implantation was the last discriminator. At the end, the tree partitioned the sample into four final nodes characterized by a single parameter or a combination of parameters (only infiltration, no infiltration and nonleft localization, no infiltration, no nonleft localization and sessile, no infiltration, no nonleft localization and no sessile implantation). The percentage correctly classified by CTA as malignant was 87.5% (Figure 4).

Table 3 Comparison of echocardiographic features between benign or malignant masses

	Total population (n = 286)	Benign CMs (n = 212)	Malignant CMs (n = 74)	P
LVEDD, mm	46.7 ± 6	47.3 ± 6.3	45.2 ± 5.7	.089
LVEDV, mL	93.3 ± 35	96 ± 38.3	85.8 ± 21.9	.182
LVEF, %	60 ± 9	60 ± 10	60 ± 7	.186
LAV, mL	51 ± 5	52 ± 4	48 ± 7	.123
DDF	13 (4.5)	9 (4.2)	4 (5.4)	.680
sysPAP, mm Hg	34 ± 15	32 ± 11	38 ± 19	.423
Mitral regurgitation				.128
Mild	160 (55.9)	123 (58)	37 (50)	
Moderate/severe	40 (14)	32 (15.1)	8 (10.8)	
Mitral stenosis				.172
Mild	8 (2.8)	8 (3.8)	0 (0)	
Moderate/severe	12 (4.2)	10 (4.7)	2 (2.7)	
Aortic regurgitation				.898
Mild	61 (21.3)	46 (21.7)	15 (20.3)	
Moderate/severe	13 (4.5)	9 (4.2)	4 (5.4)	
Aortic stenosis				.272
Mild	22 (7.7)	18 (8.5)	4 (5.4)	
Moderate/severe	5 (1.7)	5 (2.4)	0 (0)	
Tricuspid regurgitation				.069
Mild	159 (55.6)	119 (56.1)	40 (54.1)	
Moderate/severe	26 (9.1)	15 (6.6)	11 (16.2)	
Pericardial effusion	63 (22)	19 (8.9)	44 (59.5)	<.001
Mild	34 (53.9)	16 (84.2)	18 (40.9)	
Moderate/severe	29 (46.1)	3 (15.8)	26 (59.1)	
Infiltration	57 (19.9)	10 (4.7)	47 (63.5)	.001
Implantation				<.001
Lateral/superior atrial wall	57 (19.9)	31 (14.6)	26 (35.1)	
IAS	113 (39.5)	109 (51.4)	4 (5.4)	
Maximum CM diameter, mm	36 ± 19.6	30.8 ± 16	50.6 ± 21.6	<.001
Diameter > 30 mm	150 (52.4)	87 (41)	63 (85.1)	<.001
Inhomogeneity	73 (25.5)	31 (14.6)	42 (56.8)	<.001
Irregular margins	72 (25.2)	30 (14.2)	42 (56.8)	<.001
Mobility	143 (50)	130 (61.3)	13 (17.6)	<.001
Sessile mass	129 (45.1)	79 (37.3)	50 (67.6)	<.001
Polylobate mass	72 (25.2)	26 (12.3)	46 (62.2)	.001

DDF, Diastolic dysfunction; IAS, interatrial septum; LAV, left atrial volume; LVEDD, LV end-diastolic diameter; LVEDV, LV end-diastolic volume; LVEF, LV ejection fraction; sysPAP, systolic pulmonary artery pressure.

Continuous variables are expressed as mean ± SD and categorical variables as number (percentage).

Echocardiographic Diagnostic Accuracy

Complete agreement between the observer assignment and CM histology was reached in 257 of 286 cases (89.9%), at 91.5% and 85.1%

for benign and malignant CMs, respectively. The accuracy of diagnostic echocardiography was 90%, with 94% sensitivity, 80% specificity, and 92% and 84% positive and negative predictive value, respectively (Supplemental Table 2). Compared with second-level imaging techniques, echocardiography showed the best agreement with CMR (82%; diagnostic accuracy = 87%, Cohen's κ = 0.72) in predicting CMs' nature, while agreement with CCT reached 79.5% (diagnostic accuracy = 80%, Cohen's κ = 0.59), as reported in Supplemental Table 3.

Prognostic Validity of the Echocardiographic Parameters Associated with Malignancy

Of the 286 patients followed for a median of 22 months, 83 (29%) died. As expected, most deaths were observed among patients with malignancies (51 of 83 [61.4%]). The echocardiographic unweighted count was associated with Kaplan-Meier curve analysis in order to determine its ability to correlate with patient survival. Patients were classified into two groups: those with three or more of the features infiltration, polylobate shape, moderate to severe pericardial effusion, inhomogeneity, sessile implantation, and nonleft localization versus those with fewer than three features. All-cause death was significantly higher for patients with three or more echocardiographic features of malignancy compared with those with fewer than three echocardiographic features (log-rank test = 77.2, P < .001; hazard ratio, 6.65; 95% CI, 4.10-10.80; Figure 5).

DISCUSSION

The aim of this study was to describing the echocardiographic parameters of a large population of histologically confirmed CMs and look for practical clues to improve the diagnostic accuracy of malignancy. In the complex approach to CMs, the noninvasive echocardiographic approach may help simplify and optimize the subsequent diagnostic process with the support of second-level noninvasive imaging techniques (i.e., CMR, CCT, and positron emission tomography) when needed.

In summary, our main findings are as follows: (1) some echocardiographic features are strongly related to malignancies, namely, pericardial effusion, infiltration, site of implantation, inhomogeneity, sessile implantation, and polylobate shape; (2) combining at least three parameters identified as independent predictors of malignancy, we were able to increase the accuracy of echocardiographic diagnostic to approximately 90%; (3) the predictive ability of the unweighted count was very high (AUC > 0.90) and similar compared with the previously proposed weighted score; (4) CTA generated a stepwise algorithm that correctly classified malignancies in 87.5% of cases; and (5) three or more "malignant" echocardiographic parameters were associated with progressively higher risk for death.

Echocardiography in the Complex Approach to CMs

The advances in radiologic techniques achieved over the past decades have significantly increased diagnostic accuracy for CMs. In particular, the widespread use of CMR, eventually integrated with CCT and positron emission tomography depending on the specific clinical scenario, is a powerful tool in CM assessment.^{12,13,29} Nevertheless, standardized protocols for each technique in this specific context are still unavailable, and a shared stepwise diagnostic approach has yet to be defined. As a result, echocardiography, which represents the first

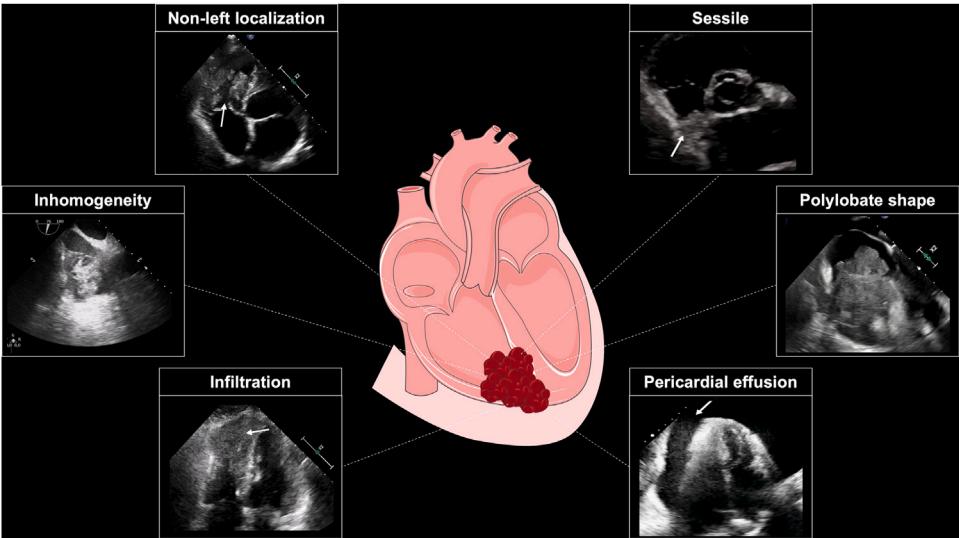


Figure 1 Echocardiographic features associated with malignant CMs. Right ventricular infiltration of a urologic metastasis. Polylobate shape of a hepatocellular carcinoma metastasis. Moderate to severe pericardial effusion of a mesothelioma metastasis. Inhomogeneity of a mediastinal lymphoma. Sessile implantation of a renal cell carcinoma metastasis involving the right atrium and the vena cava. Non-left side localization of primitive cardiac angiosarcoma.

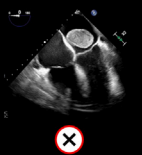
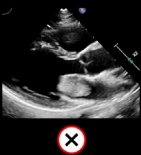
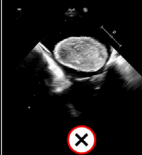
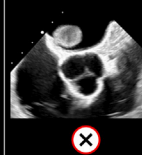
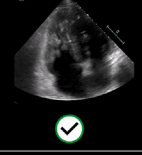
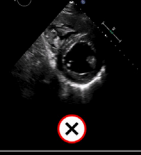
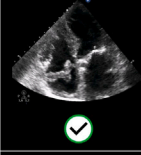
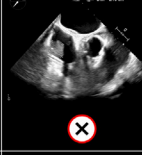
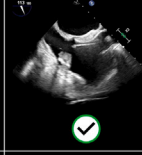
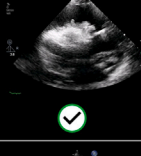
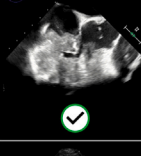
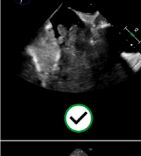
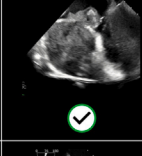
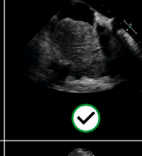
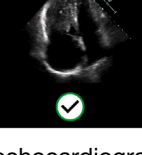
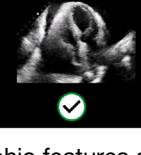
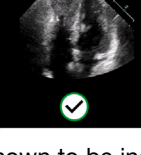
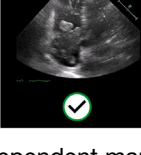
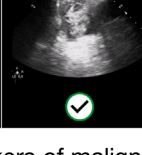
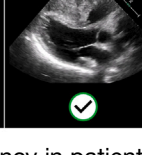
	Non-left localization	Pericardial Effusion (≥ moderate)	Infiltration	Polylobate shape	Inhomogeneity	Sessile
Primary benign tumor (e.g. myxoma)						
Pseudotumor (e.g. thrombus)						
Primary malignant tumor (e.g. angiosarcoma)						
Metastasis (e.g. melanoma, chondrosarcoma)						

Figure 2 Case examples of echocardiographic features shown to be independent markers of malignancy in patients with CMs. In each column, we report examples of the six echocardiographic parameters shown to be independent markers of malignancy in patients with CMs (non-left side localization, moderate to severe pericardial effusion, infiltration, polylobate shape, inhomogeneity, and sessile implantation). The red circle with the cross and the green circle with the check mark suggest the absence and presence, respectively, of the parameter in the selected case. Each row describes examples of echocardiographic features related to the four different histology of CMs. In the first line, we report examples of left atrial myxoma, as a primary benign tumor. In the second line, we show some cases of thrombi, as examples of cardiac pseudotumors. In the third line, two cases of cardiac angiosarcoma are shown as primary malignant tumors. In the last line, we report cases of cardiac metastasis from melanoma and chondrosarcoma as examples of metastasis. Note that images on the same line are not always from the same patient.

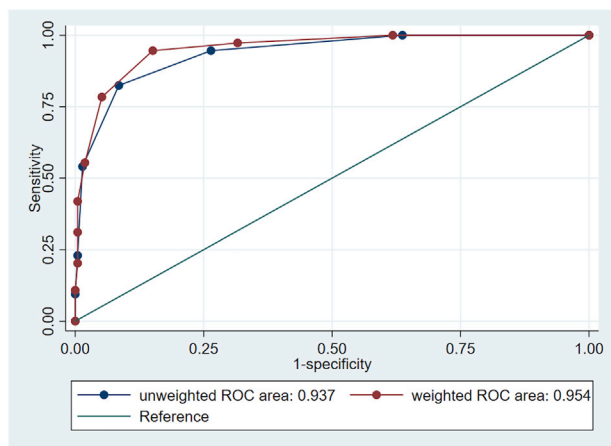


Figure 3 Receiver operating characteristic (ROC) curves showing the diagnostic performance of weighted and unweighted scores, including six echocardiographic features.

imaging step in diagnostic workup, still maintains its key role in the evaluation of CM and in the identification of patients who can benefit from second-level imaging techniques. In fact, echocardiography provides accurate information on the localization, size, shape, and surface characteristics, as well as their relationship with adjacent structures, without exposing patients to radiation.^{9,10,30} However, in this setting, scant data are available on the diagnostic accuracy of echocardiography, all deriving from retrospective studies with small sample sizes, in which only one histologic type was often included.⁹⁻¹¹ Furthermore, echocardiographic assessment is inevitably influenced by the acoustic window and the localization and histology of the mass. For instance, our registry included six patients with pulmonary artery angiosarcomas, in which the localization only in the pulmonary artery strongly limits echocardiographic diagnosis.

Score-Based and CTA-Based Diagnostic Approach

This study represents an extension of our previous work in which we proposed a weighted score that significantly increased the diagnostic yield.¹⁹ In the present analysis with additional cases, we confirmed that six echocardiographic parameters significantly discriminated between benign and malignant CMs: infiltration, moderate to severe pericardial effusion, polylobate shape, sessile implantation, inhomogeneity, and nonleft localization, which were found to be associated with malignancies (Figure 1).¹⁹ Among these variables, infiltration proved to be most associated with malignancy. Pericardial effusion was evident in 63 cases (19 [8.9%] benign masses and 44 [59.5%] of the malignant ones), and specifically, moderate to severe pericardial effusions were present in only three benign masses and 26 malignant ones (59.1%). Although this finding was not surprising, it is our opinion that the degree of pericardial effusion rather than the simple presence or absence should be considered a marker of malignancy. It should be underscored that mass size was not selected in our model, because even benign masses, such as myxomas, can reach large dimensions, remaining asymptomatic for a long time. With the goal of conveying a practical and immediately applicable clinical message, we summed the malignant echocardiographic markers and found that the coexistence of at least three parameters resulted in a very high diagnostic accuracy in predicting malignancies, noninferior to the performance exhibited by the weighted score. Indeed, the possibility of simply adding echocardiographic abnormalities encountered

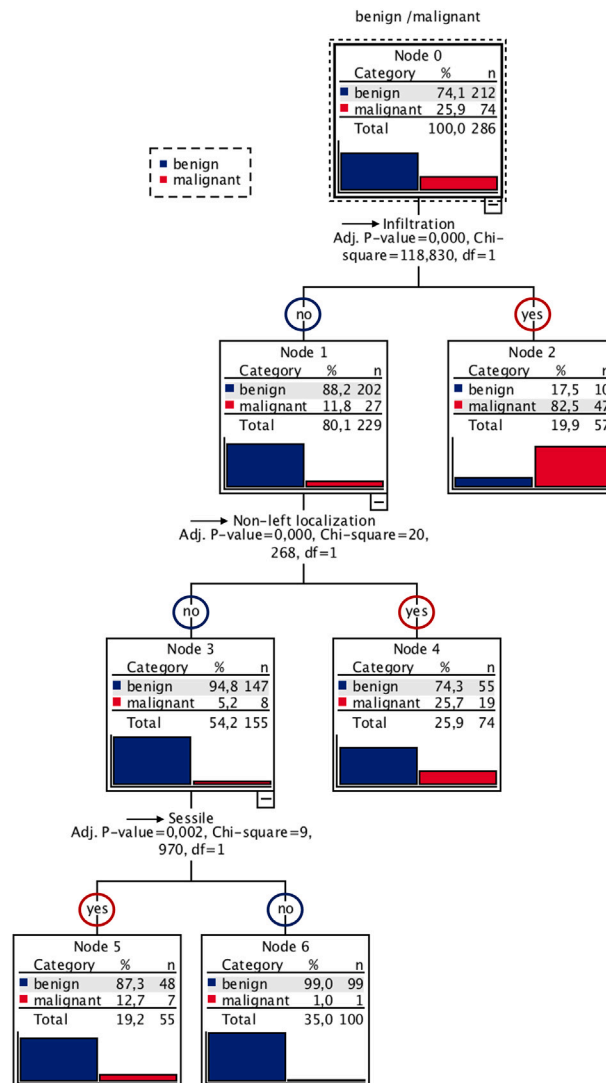


Figure 4 CTA to determine the ability of echocardiographic parameters to discriminate subgroups of patients with differential risk for malignancy.

and obtaining such an excellent diagnostic tool represents the main novelty of our work and may actually improve current clinical practice. From a practical point of view, when performing echocardiography in patients with CMs, the presence of at least three specific echocardiographic features among the six “red flags” should rapidly raise the suspicion of malignancy. On the other hand, the presence of just one or two of these parameters suggests a high likelihood of a benign mass.

Another interesting aspect of our work is that through CTA, we suggested a step-based approach that should be kept in mind when performing echocardiography in cases of suspected CM. Our results emphasize the role of infiltration, the marker most strongly associated with malignancy, even though it is not easily detectable. If noticed, infiltration alone strongly suggests malignancy; on the other hand, when infiltration is absent, or not recognizable, the operator should focus on localization, and evidence of a non-left-located mass should rapidly raise the suspicion of malignancy. The last key discriminator in this process is represented by sessile masses that, even when located

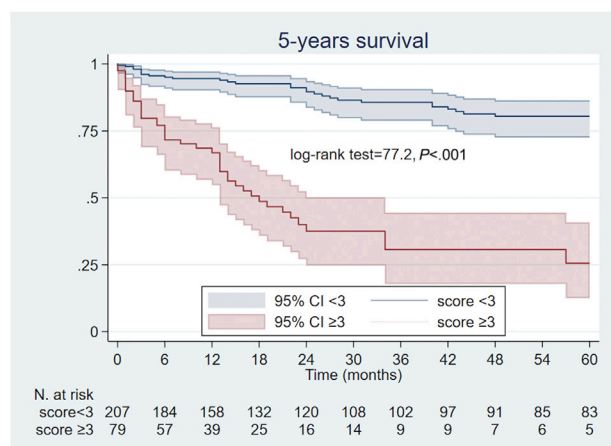


Figure 5 Kaplan-Meier survival estimates. Kaplan-Meier survival estimates (all-cause death) for patients with fewer than three versus three or more echocardiographic features associated with malignancy.

in the left heart chambers and without evident infiltration, should still represent a marker for malignancy.

Clinical Implications

Altogether, our findings may contribute to improving the echocardiographic assessment of CMs. In fact, knowing the echocardiographic malignant markers that should be looked for when incidentally encountering a CM and the possibility of reaching an extremely high diagnostic accuracy through the simple count of six parameters surely represents an evidence-based and easily applicable aid in this complex clinical scenario. Wide use of our selected echocardiographic features could help clinicians better select the proper diagnostic and therapeutic approach for each patient, also meeting the current need to optimize resources. Currently, the option for the appropriate technique is based on expert knowledge, cost/benefit ratio, and, most important, availability.³¹ Another essential advantage of our model is that patients with fewer than three echocardiographic features—which strongly suggest a benign mass—could receive rapid therapeutic and surgical management after proper evaluation, reducing the average diagnosis time and the risk for complications. In fact, the prognosis of patients with CMs does not necessarily correspond to the histology of the lesions, because even benign masses can embolize systemically and/or neurologically, producing severe disability. Last, we observed that our echocardiographic features revealed crucial prognostic information. As expected, an increase in the number of these echocardiographic parameters was associated with a progressively higher risk for death. Overall, we believe that our findings may positively affect the field of CM, providing useful clues that can substantially enhance echocardiographic assessment and quickly suggest the masses' nature, thereby minimizing the diagnostic pathway and the reliance on advanced radiologic techniques, with the goal of delivering the proper treatment in the shortest time possible.

Study Limitations

Despite providing the largest series on CMs with histologic documentation, our study had some limitations. First, the study was conducted

at a single institution. Second, surgical techniques and diagnostic procedures evolved in the 18 years of patient recruitment, so it is possible that the identification and survival of CMs changed over time.

Third, in our cohort, the prevalence of malignant cardiac tumors might have been underestimated; in fact, patients with advanced neoplasia often do not undergo further diagnostic investigations, thus precluding a histopathologic confirmation. Fourth, contrast echocardiography could be very useful in this specific setting, representing an additional valuable tool. However, we did not use ultrasound-enhancing agents. Not all facilities worldwide have access to echocardiographic contrast; the purpose of our work was to underscore the practical echocardiographic features of CMs that can be easily detected during standard echocardiography, which is widely available.

Fifth, so far, the performance of the score lacks validation in an independent external population.

CONCLUSION

Our study in a large cohort of patients with histologically proven CMs confirmed that certain echocardiographic features are strongly related to malignancy. Most important, we found that the simple addition of specific parameters improves the diagnostic accuracy and the correct identification of the masses' nature. Furthermore, a step-based approach using three characteristics provides improved discrimination in the echocardiographic evaluation of such patients. The identification of these easily identifiable echocardiographic markers may assist with initial evaluation and streamline the selection of cases that may benefit from second-level imaging evaluations.

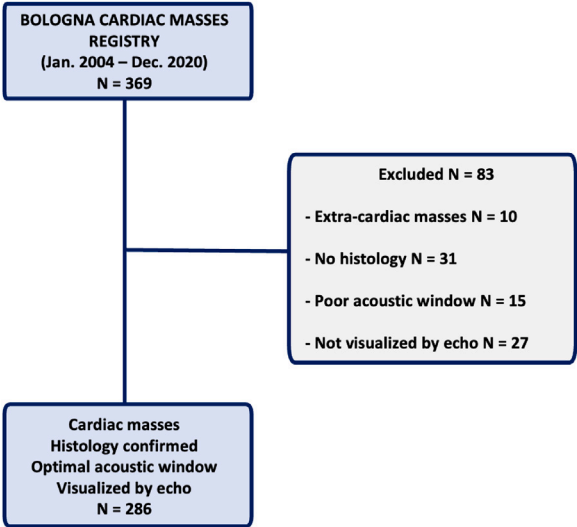
SUPPLEMENTARY DATA

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.echo.2022.12.022>.

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Supplemental Figure 1 Study flowchart.

Supplemental Table 1 Univariate and multivariable logistic regression model showing the variables independently associated with malignancy

	Univariate analysis					Multivariable analysis				
	β regression coefficient	Standard error	P	Odds ratio	95% CI	β regression coefficient	Standard error	P	Odds ratio	95% CI
Infiltration	3.560	0.404	<.001	35.163	15.93-77.63	3.316	0.578	<.001	27.551	8.88-85.51
Polylobate mass	2.464	0.308	<.001	11.753	6.30-21.93	1.890	0.493	<.001	6.620	2.520-17.39
Pericardial effusion*	3.631	0.630	<.001	37.736	10.97-129.83	2.780	0.848	.001	16.11	3.05-84.88
Sessile	1.254	0.286	<.001	3.507	2.00-6.14	1.407	0.490	.004	4.085	1.56-10.66
Inhomogeneous	2.036	0.305	<.001	7.663	4.22-13.92	1.731	0.467	<.001	5.647	2.26-14.102
Nonleft mass localization	2.572	0.350	<.001	13.089	6.59-26.01	1.380	0.477	.004	3.974	1.55-10.12

*Moderate to severe.

Supplemental Table 2 Accuracy of the observer in the detection of CM malignancy on the basis of echocardiographic parameters

	Total population of CMs (n = 286)	Number of correct readings/total benign CMs (n = 212)	Number of correct readings/total malignant CMs (n = 74)	Sensitivity, %	Specificity, %	PPV, %	NPV, %	Accuracy, %
Cardiac imager	257 (89.9%)	194 (91.5%)	63 (85.1%)	94	80	92	84	90

NPV, Negative predictive value; PPV, positive predictive value.

Supplemental Table 3 Comparison among echocardiography, CCT and CMR for assessment of malignancy in patients with CMs

	Total population of CMs (n = 286)	Number of agreements/total benign CMs (n = 212)	Number of agreements/total malignant CMs (n = 74)	Sensitivity, %	Specificity, %	PPV, %	NPV, %	Accuracy, %	Cohen's κ
Echo>3 vs CCT	132/166* (79.5%)	83/96 (86.5%)	49/70 (70%)	88	68	78	82	80	0.59
Echo>3 vs CMR	117/143† (82%)	77/87 (88.5%)	45/56 (80.4%)	87	88	93	78	87	0.72

Echo>3, Three or more parameters on echocardiography; NPV, negative predictive value; PPV, positive predictive value.

*Total number of patients with echocardiography and CCT.

†Total number of patients with echocardiography and CMR.